

Assignment 5

1. (a) Let $f : X \rightarrow \mathbb{R}$ be measurable with $f \geq 0$. If $\int_X f d\mu = 0$ then prove that $f = 0$ almost everywhere on X .

Proof. Let $f : X \rightarrow \mathbb{R}$ be measurable with $f \geq 0$, and $\int_X f d\mu = 0$. Now, suppose for sake of contradiction that f is not 0 almost everywhere on X . That is, there exists $E \subseteq X$ with $f|_E \geq \varepsilon > 0$ and $\mu(E) > 0$. Therefore $0 < \varepsilon \cdot \mu(E) \leq \int_E f d\mu$. Using countable additivity we have:

$$\int_X f d\mu = \int_E f d\mu + \int_{X \setminus E} f d\mu > 0.$$

Because $\int_{X \setminus E} f d\mu \geq 0$ by virtue of $f \geq 0$ on X . This is a contradiction, as we assumed that $\int_X f d\mu = 0$. Therefore there are no sets with non-zero measure where f is not equal to 0. In other words, $f = 0$ almost everywhere on X . □

- (b) If $f, g : X \rightarrow \mathbb{R}$ are measurable and $f(x) = g(x)$ for almost every $x \in X$ then prove that

$$\int_X f d\mu = \int_X g d\mu$$

Proof. Let $f, g : X \rightarrow \mathbb{R}$ be measurable and $f(x) = g(x)$ almost everywhere on X . Define $h : X \rightarrow \mathbb{R}$ by $h : x \mapsto |f(x) - g(x)|$. Thus, $h = 0$ almost everywhere on X , and $h \geq 0$. Hence, there must exist a subset $E \subset X$ on which $h(x) \neq 0$, and $\mu(E) = 0$. Thus, for all $x \in X \setminus E$, $h(x) = 0$. So,

$$\begin{aligned} \int_X h d\mu &= \int_{X \setminus E} h d\mu + \int_E h d\mu \leq \int_{X \setminus E} h d\mu + \int_E \sup_E \{h\} d\mu. \\ &= \int_{X \setminus E} 0 d\mu + \sup_E \{h\} \mu(E) = 0. \end{aligned}$$

It follows that:

$$\begin{aligned} \left| \int_X [f - g] d\mu \right| &\leq \int_X |f - g| d\mu \implies \left| \int_X f d\mu - \int_X g d\mu \right| \leq 0. \\ &\implies \int_X f d\mu - \int_X g d\mu = 0. \\ &\iff \int_X f d\mu = \int_X g d\mu. \end{aligned}$$

□

2. Let $(X, \mathcal{S}, \mu)$ be a measure space with f measurable on X and $\int_X |f| d\mu < \infty$. Prove that

(a) If $\{E_n\}_{n=1}^\infty$ is a sequence of measurable sets in $\mathcal{S}$ with $E_1 \subseteq E_2 \subseteq E_3 \dots$ then

$$\int_{\bigcup_{n=1}^\infty E_n} f d\mu = \lim_{n \rightarrow \infty} \int_{E_n} f d\mu$$

Proof. Let $\{T_n\} = E_n \setminus \bigcup_{k=1}^{n-1} E_k$, and $T_1 = E_1$. Thus, $T_2 = E_2 \setminus E_1$, $T_3 = E_3 \setminus (E_1 \cup E_2)$. Hence, $\{T_n\}$ is a sequence of disjoint sets such that $\bigcup_{n=1}^\infty T_n = \bigcup_{n=1}^\infty E_n$.

$$\begin{aligned} \int_{\bigcup_{n=1}^\infty E_n} f d\mu &= \int_{\bigcup_{n=1}^\infty T_n} f d\mu = \sum_{n=1}^\infty \int_{T_n} f d\mu. \\ &= \lim_{k \rightarrow \infty} \sum_{n=1}^k \int_{T_n} f d\mu. \\ &= \lim_{k \rightarrow \infty} \int_{\bigcup_{n=1}^k T_n} f d\mu = \lim_{k \rightarrow \infty} \int_{E_k} f d\mu. \end{aligned}$$

□

(b) If $\{E_n\}_{n=1}^\infty$ is a sequence of measurable sets in $\mathcal{S}$ with $E_1 \supseteq E_2 \supseteq E_3 \dots$ then

$$\int_{\bigcap_{n=1}^\infty E_n} f d\mu = \lim_{n \rightarrow \infty} \int_{E_n} f d\mu$$

Proof. We define the function $f^*(x) = \begin{cases} f(x) & \text{if } x \in \bigcap_{n=1}^\infty E_n \\ 0 & \text{if } x \notin \bigcap_{n=1}^\infty E_n \end{cases}$. Thus:

$$\int_{\bigcap_{n=1}^\infty E_n} f d\mu = \int_{E_1} f^* d\mu.$$

Define the sequence $\{f_n\} = \begin{cases} f(x) & \text{if } x \in \bigcap_{k=1}^n E_k \\ 0 & \text{if } x \notin \bigcap_{k=1}^n E_k \end{cases}$. Therefore clearly $\{f_n\} \rightarrow$

f^* pointwise. Furthermore, notice that $f_n \leq g = |f|$ and $\int_X |f| d\mu < \infty$. Apply the Dominated Convergence Theorem to yield the following:

$$\begin{aligned} \int_{\bigcap_{n=1}^\infty E_n} f d\mu &= \int_{E_1} f^* d\mu = \int_{E_1} \lim_{n \rightarrow \infty} f_n d\mu. \\ &= \lim_{n \rightarrow \infty} \int_{E_1} f_n d\mu \text{ (By DCT).} \\ &= \lim_{n \rightarrow \infty} \int_{E_n} f d\mu \text{ (By definition of } f_n). \end{aligned}$$

□

3. Evaluate $\lim_{n \rightarrow \infty} \int_0^{\frac{3}{n}} n \cos(t^3) dt$.

Let $u = nt$:

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_0^{\frac{3}{n}} n \cos(t^3) dt &= \lim_{n \rightarrow \infty} \int_0^3 n \cos\left(\frac{u^3}{n^3}\right) \frac{du}{n} \\ &= \lim_{n \rightarrow \infty} \int_0^3 \cos\left(\frac{u^3}{n^3}\right) du. \end{aligned}$$

Notice that $\mu([0, 3]) = 3 < \infty$ and that $|\cos(\frac{u^3}{n^3})| \leq 1$ for all $u \in [0, 3]$, and all $n \in \mathbb{N}$. Thus, we can apply the bounded convergence theorem:

$$\lim_{n \rightarrow \infty} \int_0^3 \cos\left(\frac{u^3}{n^3}\right) du = \int_0^3 \lim_{n \rightarrow \infty} \cos\left(\frac{u^3}{n^3}\right) du = \int_0^3 1 \cdot du = 3.$$

4. Consider the space $(\mathbb{N}, \mathcal{P}(\mathbb{N}), \mu)$ where $\mu(A) = \begin{cases} n & \text{if } A \text{ has size } n \\ \infty & \text{if } A \text{ is infinite} \end{cases}$ is counting measure.

- (a) Let $f : \mathbb{N} \rightarrow \mathbb{R}$ with $f \geq 0$. Compute $\int f d\mu$ (Hint: Start with simple functions)

Let $\varphi : \mathbb{N} \rightarrow \mathbb{R}$ be simple. Thus, $\varphi = \sum_{k=1}^n c_k \cdot \chi_{E_k}$, where each E_k are disjoint subsets of $\mathbb{N}$. Because φ is simple,

$$\int_E \varphi d\mu = \sum_{k=1}^n c_k \cdot \mu(E_k).$$

We further break up φ into the subsets $E_k = \{k\}$, giving:

$$\int_E \varphi d\mu = \sum_{k=1}^{\infty} \varphi(k) \cdot \mu(\{k\}) = \sum_{k=1}^{\infty} \varphi(k) \cdot 1 = \varphi(1) + \varphi(2) + \dots$$

Note that this is not reliant on φ being simple, thus:

$$\int_E f d\mu = \sum_{k=1}^{\infty} f(k).$$

- (b) Let $f \in \mathcal{L}(\mu)$ and compute $\|f\|_1$.

Proof. Let $f \in \mathcal{L}(\mu)$, thus computing $\|f\|_1$:

$$\|f\|_1 = \int_{\mathbb{N}} |f| d\mu = |f(1)| + |f(2)| + \dots = \sum_{k=1}^{\infty} |f(k)|.$$

□

5. Suppose $(X, \mathcal{S}, \mu)$ is a measure space with $\mu(X) = 1$ and $h \in \mathcal{L}^1(X)$. Then for $c > 0$ prove that

$$\mu\left(\left\{x \in X : \left|h(x) - \int_X h d\mu\right| \geq c\right\}\right) \leq \frac{1}{c^2} \left(\int_X h^2 d\mu - \left(\int_X h d\mu\right)^2\right)$$

Proof. let $(X, \mathcal{S}, \mu)$ be a measure space with $\mu(X) = 1$ and let $h : X \rightarrow \mathbb{R}$ with $h \in \mathcal{L}^1(X)$. Consider the function f defined by $f(x) = (h(x) - \int_X h d\mu)^2$. We apply Markov's inequality to get the following:

$$\mu(\{x \in X : |f(x)| \geq k\}) \leq \frac{1}{k} \|f\|_1.$$

Note that $f \in \mathcal{L}^1(X)$ because $h \in \mathcal{L}^1(X)$ and so $\int_X |f| d\mu = \int_X |h - \int_X h d\mu|^2 d\mu < \infty$. Thus, f is integrable. So:

$$\begin{aligned} \mu(\{x \in X : |f(x)| \geq k\}) &\leq \frac{1}{k} \|f\|_1. \\ \implies \mu(\{x \in X : |h(x) - \int_X h d\mu|^2 \geq k\}) &\leq \frac{1}{k} \|[h - \int_X h d\mu]^2\|_1. \\ \implies \mu(\{x \in X : |h(x) - \int_X h d\mu| \geq \sqrt{k}\}) &\leq \frac{1}{k} \|[h - \int_X h d\mu]^2\|_1. \\ &\leq \frac{1}{k} \cdot \int_X [h^2 - (\int_X h d\mu)^2] d\mu. \\ &= \frac{1}{k} \cdot [\int_X h^2 d\mu - \int_X (\int_X h d\mu)^2 d\mu]. \end{aligned}$$

Because $\mu(X) = 1$ and $(\int_X h d\mu)^2$ is a constant, we have that:

$$\begin{aligned} \int_X (\int_X h d\mu)^2 d\mu &= \mu(X) \cdot (\int_X h d\mu)^2 \\ &= 1 \cdot (\int_X h d\mu)^2. \\ &= (\int_X h d\mu)^2. \end{aligned}$$

So finally, letting $c = \sqrt{k}$, we have:

$$\mu\left(\left\{x \in X : \left|h(x) - \int_X h d\mu\right| \geq c\right\}\right) \leq \frac{1}{c^2} \left(\int_X h^2 d\mu - \left(\int_X h d\mu\right)^2\right).$$

as desired. □

6. Use the Borel-Cantelli lemma to finish the proof from class that if f_n converges in measure to f then there is a subsequence of f_n that converges almost everywhere to f pointwise.

Proof. For every $k \in \mathbb{N}$, there exists $n_k \in \mathbb{N}$ s.t. if $j \geq n_k$, then:

$$\mu(\{x \in X : |f(x) - f_j(x)| > \frac{1}{k}\}) < \frac{1}{2^k}.$$

Let $E_k = \{x \in X : |f_{n_k}(x) - f(x)| > \frac{1}{k}\}$. Then $\mu(E_k) < \frac{1}{2^k}$. Notice that $\sum_{k=1}^{\infty} \mu(E_k) = 1 < \infty$. We apply the Borel-Cantelli lemma to get $\mu(\limsup_{n \rightarrow \infty} E_n) = 0$. By definition of lim sup, we have:

$$\mu(\bigcap_{k \geq 1} \bigcup_{n \geq k} E_n) = \mu(\limsup_{n \rightarrow \infty} E_n) = 0.$$

Taking complements and excessively using DeMorgan's Laws..

$$\begin{aligned} \mu([\bigcap_{k \geq 1} \bigcup_{n \geq k} E_n]^C) &= \mu(\bigcup_{k \geq 1} [\bigcup_{n \geq k} E_n]^C) \\ &= \mu(\bigcup_{k \geq 1} \bigcap_{n \geq k} E_n^C) \\ &= \mu(\bigcup_{k \geq 1} \bigcap_{n \geq k} \{x \in X : |f_{n_k}(x) - f(x)| \leq \frac{1}{k}\}). \end{aligned}$$

Note that because $\bigcap_{k \geq 1} \bigcup_{n \geq k} E_n$ and $[\bigcap_{k \geq 1} \bigcup_{n \geq k} E_n]^C$ are disjoint, the sum of their measures must be equal to $\mu(X)$. Thus, because $\mu(\bigcap_{k \geq 1} \bigcup_{n \geq k} E_n) = 0$, it follows that:

$$\mu([\bigcap_{k \geq 1} \bigcup_{n \geq k} E_n]^C) = \mu(\bigcup_{k \geq 1} \bigcap_{n \geq k} \{x \in X : |f_{n_k}(x) - f(x)| \leq \frac{1}{k}\}) = \mu(X).$$

This is equivalent to pointwise convergence of the subsequence $\{f_{n_k}\}$ by question 1 of homework 4.

□